Week 10

Real-Estate Analytics Dashboard-Report

(Power BI Desktop)

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1. Introduction

This week I built a real-estate dashboard in Power BI using the King County House Sales dataset that I cleaned in Week 9. The idea is to help a real-estate manager easily see how house prices change with location and house features, and how sales look overall.

2. Objectives

- Create the main DAX measures: Total Properties Sold, Average House Price, Median Price, Average Living Area and Average Price per SqFt.
- Show the houses on a map using their latitude and longitude.
- Check how grade, condition and waterfront affect the price.
- Add range filters so the dashboard can be filtered by price and living area.

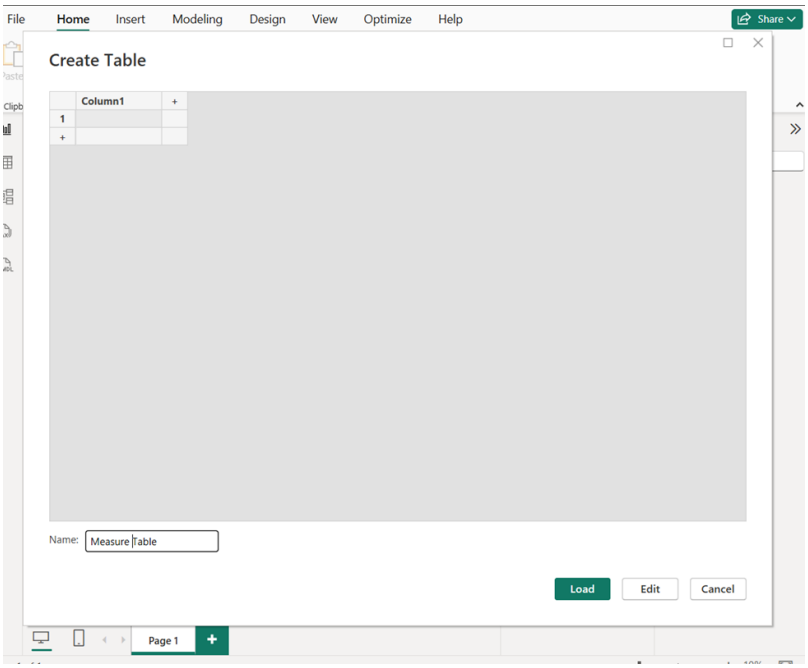
3. Dataset

I used the King County house sales data (kc_house_data), which I already preprocessed in Week 9. It has columns like price, bedrooms, bathrooms, sqft_living, grade, condition, waterfront, latitude and longitude. I also used the new columns I added earlier, such as Price per SqFt, Property Size and Price Tier.

4. Steps I Followed

Step 1: Creating a Measure Table

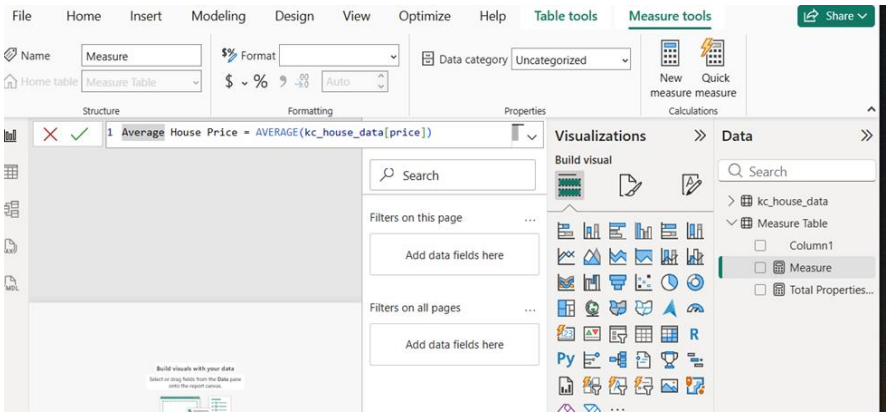
First I made an empty table named "Measure Table" using the Enter Data option. I keep all my DAX measures inside it so they stay in one place and are easy to find.



Step 2: Writing the DAX Measures

Next I created the measures one by one using New Measure. The first one was Average House Price, which takes the average of the price column.

```
Average House Price = AVERAGE(kc_house_data[price])
```



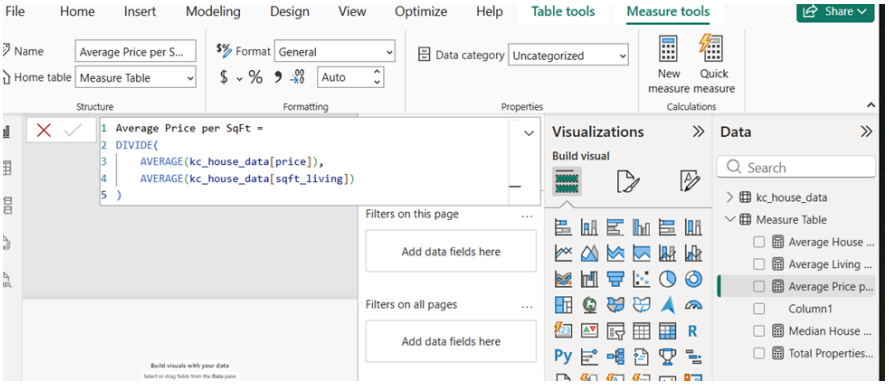
I created Total Properties Sold, Median House Price and Average Living Area in the same way.

Step 3: Average Price per SqFt

For this measure I divided the average price by the average living area. I used DIVIDE instead of the normal "/" sign because it avoids errors if the value is zero.

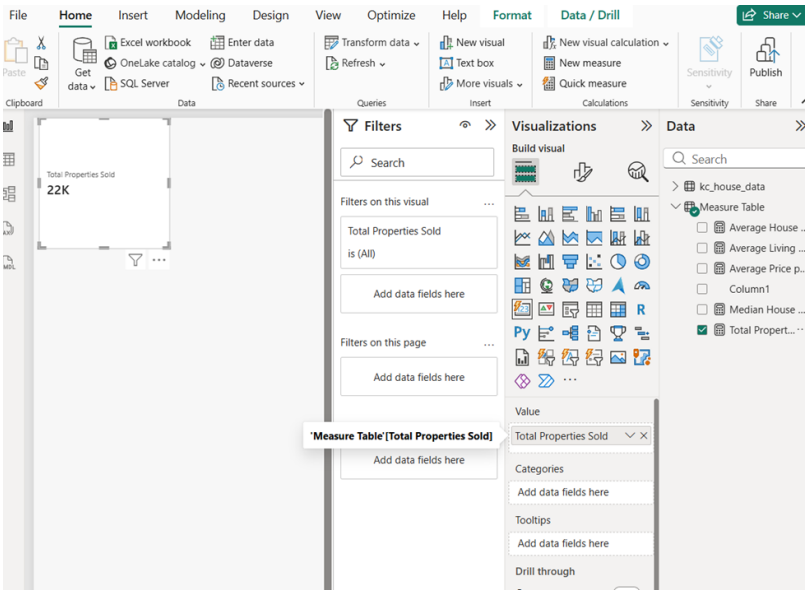
```
Average Price per SqFt =
DIVIDE(
    AVERAGE(kc_house_data[price]),
    AVERAGE(kc_house_data[sqft_living])

)
```



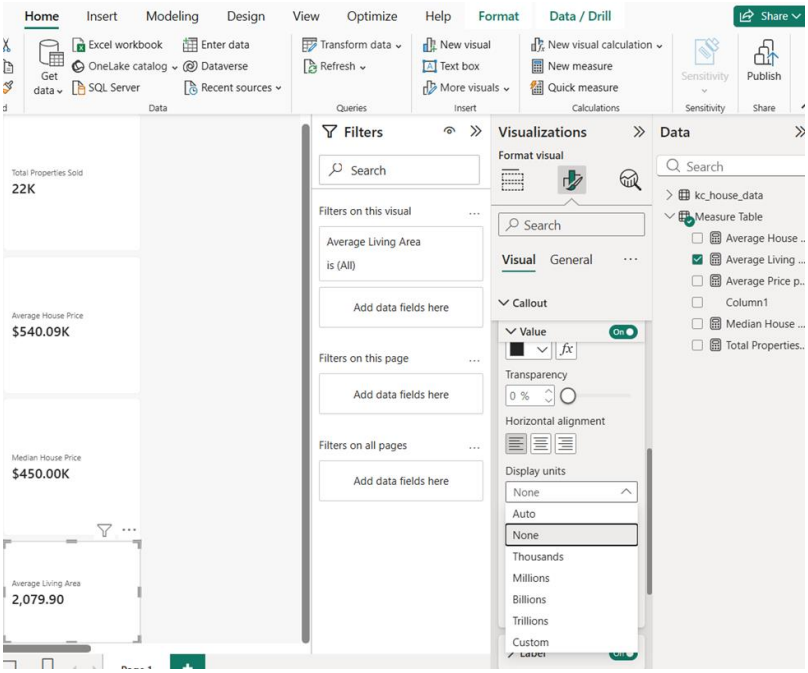
Step 4: Showing the Measures as Cards

I added a Card visual for each measure so the key numbers are visible at the top of the dashboard. For example, the Total Properties Sold card shows 22K.



Step 5: Formatting the Cards

The average living area was showing in a short form, so I changed Display units to "None" in the Format pane. This shows the full number like 2,079.90 instead of a rounded one.

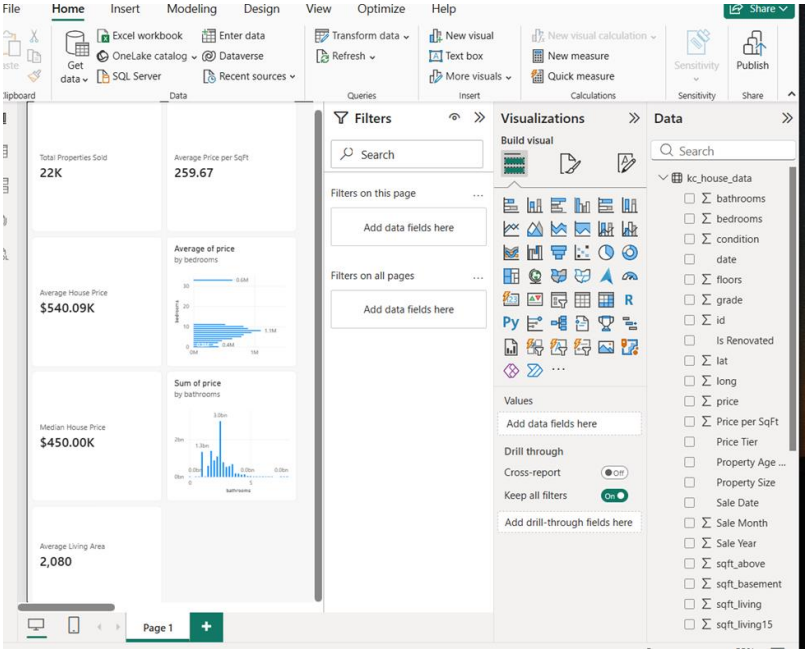


After this, I had all five cards ready.

Measure	What it shows
Total Properties Sold	Number of houses sold (about 22K)
Average House Price	Average sale price of a house
Median House Price	The middle price, so a few very costly houses don't distort it
Average Living Area	Average living space in sq ft
Average Price per SqFt	Average price divided by average living area

Step 6: Price Charts

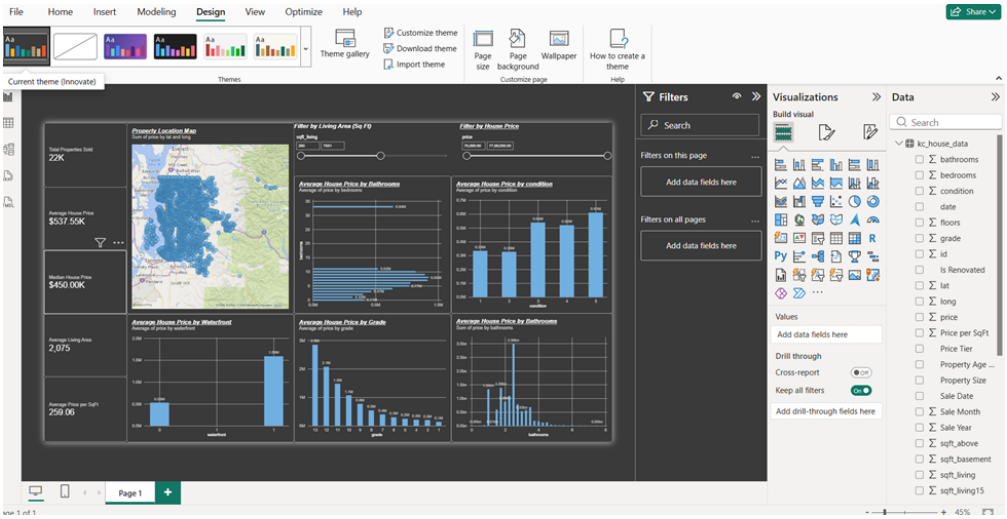
Then I added charts to see how the price changes with house features, such as average price by bedrooms and total price by bathrooms. I put these next to the cards to check the layout.



Step 7: Final Dashboard

Finally, I applied a dark theme and arranged everything on one page. The final dashboard has:

- Five cards with the key numbers.
- A property location map made with latitude and longitude.
- Two range sliders: one for living area (sq ft) and one for house price.
- Bar charts showing average price by bedrooms, bathrooms, condition, waterfront and grade.



5. Conclusion

This week helped me understand how to build a complete dashboard from start to finish. I learned how to create DAX measures, keep them in a measure table, use maps, and add sliders to make the report interactive. The dashboard now gives a quick and clear view of house prices and sales, which is useful for a real-estate manager.